सहायक प्राध्यापक (पदार्थ विज्ञान)
महाराष्ट्र अभियांत्रिकी शिक्षक सेवा, गट-अ
उच्च व तंत्र शिक्षण विभाग
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Higher and Technical Education Department

परीक्षेचे टप्पे: १) लेखी परीक्षा - २०० गुण २) मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (१११७)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक या विषयांमध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ. क्र.	घटक व उपघटक
1	**Section 1: Mathematical Physics**
	Vector calculus: linear vector space, basis, orthogonality and completeness; matrices; similarity transformations, diagonalization, eigen values and eigen vectors; linear differential equations: second order linear differential equations and solutions involving special functions; Complex analysis: Cauchy-Riemann conditions, Cauchy's theorem, singularities, residue theorem and applications; Laplace and Fourier transforms and their applications.
2	**Section 2: Classical Mechanics**
	Lagrangian formulation: D'Alembert's principle, Euler-Lagrange equation, Hamilton's principle and Hamilton's equations of motion, calculus of variations, symmetry and conservation laws; central force motion: Kepler Laws; Small oscillations, coupled oscillations, normal modes and eigen frequencies, rigid body dynamics, moment of inertia tensor, orthogonal transformations, Euler angles, Torque free motion of a symmetric top; Canonical transformations, equations of Canonical transformations; Poisson brackets and Poisson equation of motion. Special theory of relativity, Lorentz transformations, relativistic kinematics, mass-energy equivalence.
3	**Section 3: Electromagnetic Theory**
	Solutions of electrostatic and magnetostatic problems including boundary value problems. Method of images and charge distribution; dielectrics and conductors; magnetic materials, multipole expansión; Maxwell's equations; scalar and vector potentials: Coulomb and Lorentz gauges, electromagnetic waves in free space, non-conducting, and conducting media; reflection and transmission at normal and oblique incidences, polarization of electromagnetic waves; Poynting vector, Poynting theorem, energy and momentum of electromagnetic waves, Maxwell's stress tensor; radiation from a moving charge.
4	**Section 4: Quantum Mechanics**
	Postulates and development of quantum mechanics; uncertainty principle; Schrodinger equation; one dimensional potentials: step potential, finite rectangular well, tunneling from a potential barrier, particle in a box, harmonic oscillator. Two and three dimensional systems and concept of degeneracy.

	Hydrogen atom, eigen values and eigen functions, Angular momentum and spin, Pauli's spin matrices and Pauli's spinors; addition of angular momenta. Variational method and WKB approximation, time independent perturbation theory; elementary scattering theory, Born approximation; symmetries in quantum mechanical systems. Dirac Bra-Ket notation, linear vectors and operators in Hilbert space.
5	**Section 5: Thermodynamics and Statistical Physics**
	Laws of thermodynamics, equipartition theorem, calculations of specific heat of solids and gases, concept of macrostates and microstates; phase space; ensembles, partition function, free energy, calculation of thermodynamic quantities. Classical and quantum statistics: degenerate Fermi gas. Black body radiation, and Planck's distribution law; Stefan's law; Bose- Einstein condensation; first and second order phase transitions, phase equilibria, critical point, Liouville's theorem.
6	**Section 6: Atomic and Molecular Physics**
	Spectra of one-and many-electron atoms; spin-orbit interaction: LS and jj couplings; fine and hyperfine structures; Zeeman and Stark effects; electric dipole transitions and selection rules. Rotational and vibrational spectra of diatomic molecules; electronic transitions in diatomic molecules, Franck-Condon principle: Raman effect; EPR, NMR, ESR, X-ray spectra; lasers: Einstein coefficients, population inversion, two and three level systems.
7	**Section 7: Solid State Physics**
	Elements of crystallography; diffraction methods for structure determination; bonding in solids, lattice vibrations and thermal properties of solids; free electron theory; band theory of solids: nearly free electron and tight binding models. Metals, semiconductors and insulators; conductivity, mobility and the concept of effective mass. Optical properties of solids; Kramers-Kronig relation, intra- and inter-band transitions; dielectric properties of solid; dielectric function, polarizability, ferroelectricity. Magnetic properties of solids; dia, para, ferro, antiferro and ferri-magnetism, domains and magnetic anisotropy. Superconductivity: Type-l and Type II superconductors, Meissner effect, London equation, BCS Theory, flux quantization.
8	**Section 8: Electronics**
	Semiconductors in equilibrium: electron and hole statistics in intrinsic and extrinsic semiconductors; metal-semiconductor junctions; Ohmic and rectifying contacts; PN diodes, bipolar junction transistors, insulated gate bipolar transistor (Igbt), field effect transistors; MOS-FET, negative and positive feedback circuits; oscillators, operational amplifiers, active filters; basics of digital logic circuits, combinational and sequential circuits, flip-flops, timers, counters, registers, A/D and D/A conversion.
9	**Section 9: Nuclear and Particle Physics**
	Nuclear radii and charge distributions, nuclear binding energy, electric and magnetic moments, semi-empirical mass formula; nuclear models; liquid drop model, shell model, nuclear force and two nucleon problem alpha, beta, gamma decay in nuclei, electromagnetic transitions in nuclei; Rutherford scattering, nuclear reactions, fission and fusion; particle accelerators and detectors; elementary particles; photons, baryons, mesons and leptons, the quark model, isospin symmetry and conservation laws.

दिनांक – १६ ऑक्टोबर, २०२४

अवर सचिव
महाराष्ट्र लोकसेवा आयोग